

10.18 Consider the data file *mroz* on working wives. Use the 428 observations on married women who participate in the labor force. In this exercise, we examine the effectiveness of a parent's college education as an instrumental variable.

- a. Create two new variables. *MOTHERCOLL* is a dummy variable equaling one if *MOTHEREDUC* > 12, zero otherwise. Similarly, *FATHERCOLL* equals one if *FATHEREDUC* > 12 and zero otherwise. What percentage of parents have some college education in this sample?
- b. Find the correlations between *EDUC*, *MOTHERCOLL*, and *FATHERCOLL*. Are the magnitudes of these correlations important? Can you make a logical argument why *MOTHERCOLL* and *FATHERCOLL* might be better instruments than *MOTHEREDUC* and *FATHEREDUC*?
- c. Estimate the wage equation in Example 10.5 using *MOTHERCOLL* as the instrumental variable. What is the 95% interval estimate for the coefficient of *EDUC*?
- d. For the problem in part (c), estimate the first-stage equation. What is the value of the *F*-test statistic for the hypothesis that *MOTHERCOLL* has no effect on *EDUC*? Is *MOTHERCOLL* a strong instrument?
- e. Estimate the wage equation in Example 10.5 using *MOTHERCOLL* and *FATHERCOLL* as the instrumental variables. What is the 95% interval estimate for the coefficient of *EDUC*? Is it narrower or wider than the one in part (c)?
- f. For the problem in part (e), estimate the first-stage equation. Test the joint significance of *MOTHERCOLL* and *FATHERCOLL*. Do these instruments seem adequately strong?
- g. For the IV estimation in part (e), test the validity of the surplus instrument. What do you conclude?

a.

```
> cat("母親受過大學以上教育的比例:", round(pct_mother, 2), "%\n")
母親受過大學以上教育的比例: 12.15 %
> cat("父親受過大學以上教育的比例:", round(pct_father, 2), "%\n")
父親受過大學以上教育的比例: 11.68 %
```

b.

	educ	MOTHERCOLL	FATHERCOLL
educ	1.0000000	0.3594705	0.3984962
MOTHERCOLL	0.3594705	1.0000000	0.3545709
FATHERCOLL	0.3984962	0.3545709	1.0000000

表示父母的大学教育程度對於子女的受教育年次有預測力。

對於 IV3: $\text{corr}(Z_k, X_k) \neq 0$ 是有利證據。

但父母的大学教育程度對於子女的薪資沒有直接影響。符合

IV1: Z_k doesn't have direct effect on y_i .

c.

$$\ln(\text{WAGE}) = \beta_1 + \beta_2 \text{EXPER} + \beta_3 \text{EXPER}^2 + \beta_4 \text{EDUC} + e$$

c. 2SLS (MOTHERCOLL) EDUC 估計係數:

```
> print(coef(mroz1.iv)["educ"])
```

educ

```
0.07601796
```

```
> cat("c. EDUC 係數的 95% 信賴區間:\n")
```

c. EDUC 係數的 95% 信賴區間:

```
> print(confint(mroz1.iv, "educ", level = 0.95))
```

2.5 % 97.5 %

```
educ -0.001219763 0.1532557
```

d. 第一階段 F 檢定 ($MOTHERCOLL = 0$):
> print(linearHypothesis(stage1_d, "MOTHERCOLL = 0"))

Linear hypothesis test:
MOTHERCOLL = 0

Model 1: restricted model

Model 2: educ ~ MOTHERCOLL + exper + I(exper^2)

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	425	2219.2				
2	424	1929.9	1	289.32	63.563	1.455e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$H_0: MOTHERCOLL = 0$ v.s. $H_1: MOTHERCOLL \neq 0$

$F^* = 63.563 > F_{10}$ (經驗法則), 拒絕 H_0 , 表示有證據證明

MOTHERCOLL 是一個強的工具變數

e. 2SLS (MOTHERCOLL & FATHERCOLL) EDUC 估計係數:

> print(coef(mroz1.iv1)["educ"])
educ

0.08784765

> cat("e. EDUC 係數的 95% 信賴區間:\n")

e. EDUC 係數的 95% 信賴區間:

> print(confint(mroz1.iv1, "educ", level = 0.95))

2.5 % 97.5 %

educ 0.02751845 0.1481769

區間寬度 $c = 0.1549$, $e = 0.1210$, 所以 e 的區間寬度比 c 窄

信賴區間變窄是因為加入第2個有效的IV: FATHERCOLL,

進而提高了估計效率 (降低標準誤)

f. 第一階段聯合 F 檢定 (MOTHERCOLL=0 & FATHERCOLL=0):

```
> print(linearHypothesis(stage1_f, c("MOTHERCOLL = 0", "FATHERCOLL = 0")))
```

Linear hypothesis test:
MOTHERCOLL = 0
FATHERCOLL = 0

Model 1: restricted model
Model 2: educ ~ MOTHERCOLL + FATHERCOLL + exper + I(exper^2)

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	425	2219.2				
2	423	1748.3	2	470.88	56.963	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$H_0: \text{MOTHERCOLL} = \text{FATHERCOLL} = 0$

$F^* = 56.963 > F_{10}$, 拒絕 H_0 , 表示有證據證明這組工具變換是足夠強的, 對於女性的教育程度有極大的聯合解釋力。

g. Sargan Test (Overidentifying restrictions):

```
> print(summary(mroz1.iv1, diagnostics = TRUE)$diagnostics)
```

	df1	df2	statistic	p-value
Weak instruments	2	423	56.9628992	1.241993e-22
Wu-Hausman	1	423	0.5186358	4.718220e-01
Sargan	1	NA	0.2375851	6.259557e-01

$H_0: IVs \text{ are valid}$

$p\text{-value} = 0.6260 > 0.05$, 無法拒絕 H_0 , 表示 FATHERCOLL 通過了 overidentifying test, 這組工具變換是有效的。

10.20 The CAPM [see Exercises 10.14 and 2.16] says that the risk premium on security j is related to the risk premium on the market portfolio. That is

$$r_j - r_f = \alpha_j + \beta_j(r_m - r_f)$$

where r_j and r_f are the returns to security j and the risk-free rate, respectively, r_m is the return on the market portfolio, and β_j is the j th security's "beta" value. We measure the market portfolio using the Standard & Poor's value weighted index, and the risk-free rate by the 30-day LIBOR monthly rate of return. As noted in Exercise 10.14, if the market return is measured with error, then we face an errors-in-variables, or measurement error, problem.

- a. Use the observations on Microsoft in the data file *capm5* to estimate the CAPM model using OLS. How would you classify the Microsoft stock over this period? Risky or relatively safe, relative to the market portfolio?
- b. It has been suggested that it is possible to construct an IV by ranking the values of the explanatory variable and using the rank as the IV, that is, we sort $(r_m - r_f)$ from smallest to largest, and assign the values $RANK = 1, 2, \dots, 180$. Does this variable potentially satisfy the conditions IV1–IV3? Create $RANK$ and obtain the first-stage regression results. Is the coefficient of $RANK$ very significant? What is the R^2 of the first-stage regression? Can $RANK$ be regarded as a strong IV?
- c. Compute the first-stage residuals, $\hat{v}$, and add them to the CAPM model. Estimate the resulting augmented equation by OLS and test the significance of $\hat{v}$ at the 1% level of significance. Can we conclude that the market return is exogenous?
- d. Use $RANK$ as an IV and estimate the CAPM model by IV/2SLS. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?
- e. Create a new variable $POS = 1$ if the market return $(r_m - r_f)$ is positive, and zero otherwise. Obtain the first-stage regression results using both $RANK$ and POS as instrumental variables. Test the joint significance of the IV. Can we conclude that we have adequately strong IV? What is the R^2 of the first-stage regression?
- f. Carry out the Hausman test for endogeneity using the residuals from the first-stage equation in (e). Can we conclude that the market return is exogenous at the 1% level of significance?
- g. Obtain the IV/2SLS estimates of the CAPM model using $RANK$ and POS as instrumental variables. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?
- h. Obtain the IV/2SLS residuals from part (g) and use them (not an automatic command) to carry out a Sargan test for the validity of the surplus IV at the 5% level of significance.

a.

```
Call:
lm(formula = y ~ x, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27424 -0.04744 -0.00820  0.03869  0.35801

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003250   0.006036   0.538   0.591
x            1.201840   0.122152   9.839 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08083 on 178 degrees of freedom
Multiple R-squared:  0.3523,    Adjusted R-squared:  0.3486
F-statistic: 96.8 on 1 and 178 DF,  p-value: < 2.2e-16
```

$$r_j - r_f = 0.00325 + 1.20184(r_m - r_f)$$

$\beta = 1.20184 > 1$, 表示 Microsoft 的風險大於市場風險。

$\therefore$ Microsoft 應被分類為 risky 的資產。

b.

```
Call:
lm(formula = x ~ RANK, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.110497 -0.006308  0.001497  0.009433  0.029513

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -7.903e-02  2.195e-03  -36.0 <2e-16 ***
RANK         9.067e-04  2.104e-05   43.1 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01467 on 178 degrees of freedom
Multiple R-squared:  0.9126,    Adjusted R-squared:  0.9121
F-statistic: 1858 on 1 and 178 DF,  p-value: < 2.2e-16
```

Rank 係最顯著。

$$R^2 = 0.9126$$

IV₁: z_2 doesn't have direct effect on $y_2 \rightarrow$ 排名不直接影響

Microsoft 的報酬率

IV₂: $\text{Corr}(z_2, e_2) = 0 \rightarrow$ 排名和模型的測量誤差不相關

IV₃: $\text{Corr}(z_2, y_2) \neq 0 \rightarrow$ Rank 是由 $(r_m - r_f)$ 排序而來, 故與 $(r_m - r_f)$ 高度相關

因為 Rank 係取顯著, 且 $R^2 = 0.9126$. Rank 對於 $(r_m - r_f)$ 解釋力大且符合 conditions IV₁ - IV₃. $\therefore$ Rank 是 strong IV.

C.

```
Call:
lm(formula = y ~ x + v_hat, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27140 -0.04213 -0.00911  0.03423  0.34887

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003018   0.005984   0.504   0.6146
x            1.278318   0.126749  10.085 <2e-16 ***
v_hat       -0.874599   0.428626  -2.040   0.0428 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08012 on 177 degrees of freedom
Multiple R-squared:  0.3672,    Adjusted R-squared:  0.36
F-statistic: 51.34 on 2 and 177 DF,  p-value: < 2.2e-16
```

$H_0: \sigma = 0$ v.s $H_1: \sigma \neq 0$

$p\text{-value} = 0.0428 > 0.01 \rightarrow$ 無法拒絕 H_0 .

當 $\alpha = 0.01$ 時, 沒有足夠證據證明市場報酬具有內生性,
所以可以推論市場報酬具有外生性

d.

Call:

ivreg(formula = y ~ x | RANK, data = capm5)

Residuals:

	Min	1Q	Median	3Q	Max
	-0.271625	-0.049675	-0.009693	0.037683	0.355579

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.003018	0.006044	0.499	0.618
x	1.278318	0.128011	9.986	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08092 on 178 degrees of freedom

Multiple R-Squared: 0.3508, Adjusted R-squared: 0.3472

Wald test: 99.72 on 1 and 178 DF, p-value: < 2.2e-16

IV estimate = 1.278318 > OLS estimate = 1.20184

符合預期，因為市場報酬存在測量誤差，會使 OLS estimate 產生 attenuation bias，代表 OLS 會低估，所以 IV estimate 應大於 OLS estimate。

e. Call:
lm(formula = x ~ RANK + POS, data = capm5)

Residuals:

	Min	1Q	Median	3Q	Max
	-0.109182	-0.006732	0.002858	0.008936	0.026652

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.0804216	0.0022622	-35.55	<2e-16 ***
RANK	0.0009819	0.0000400	24.55	<2e-16 ***
POS	-0.0092762	0.0042156	-2.20	0.0291 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01451 on 177 degrees of freedom
Multiple R-squared: 0.9149, Adjusted R-squared: 0.9139
F-statistic: 951.3 on 2 and 177 DF, p-value: < 2.2e-16

$$R^2 = 0.9149$$

H_0 : RANK 和 POS 係數皆為 0

$F^* = 951.3 > F_{10}$, $p\text{-value} < 0.05 \rightarrow$ 拒絕 H_0

有證據支持擁有 adequately strong IV.

f. Call:
lm(formula = y ~ x + v_hat_e, data = capm5)

Residuals:

	Min	1Q	Median	3Q	Max
	-0.27132	-0.04261	-0.00812	0.03343	0.34867

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.003004	0.005972	0.503	0.6157
x	1.283118	0.126344	10.156	<2e-16 ***
v_hat_e	-0.954918	0.433062	-2.205	0.0287 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07996 on 177 degrees of freedom
Multiple R-squared: 0.3696, Adjusted R-squared: 0.3625
F-statistic: 51.88 on 2 and 177 DF, p-value: < 2.2e-16

$H_0: \sigma_e = 0$ v.s. $H_1: \sigma_e \neq 0$

$\therefore p\text{-value} = 0.0289 > 0.01 \therefore$ 無證據拒絕 H_0

當 $\alpha = 0.01$ 時, 沒有足夠證據證明市場報酬具有內生性,
所以可以推論市場報酬具有外生性

9

```
Call:
ivreg(formula = y ~ x | RANK + POS, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27168 -0.04960 -0.00983  0.03762  0.35543

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003004   0.006044   0.497    0.62
x             1.283118   0.127866  10.035 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08093 on 178 degrees of freedom
Multiple R-Squared:  0.3507,    Adjusted R-squared:  0.347
Wald test: 100.7 on 1 and 178 DF,  p-value: < 2.2e-16
```

IV estimate = 1.283118 > OLS estimate = 1.20184

符合預期, 因為市場報酬存在測量誤差, 會使 OLS estimate
產生 attenuation bias, 代表 OLS 會低估, 所以 IV estimate
應大於 OLS estimate.

h.

```
> cat("Sargan Statistic (nR^2):", sargan_stat, "\n")  
Sargan Statistic (nR^2): 0.5584634  
> cat("P-value:", p_val, "\n")  
P-value: 0.45488
```

H_0 : IVs are valid v.s H_1 : IVs are invalid

$\because P\text{-value} = 0.45488 > 0.05 \therefore$ 無法拒絕 H_0

可以推論 多餘的 IV 並未與誤差項產生顯著相關。

(沒有證據證明 POS 違反外生性條件)

因此, 在 $\alpha = 0.05$, 額外的 IV 是有效的。